

# Math GRE Prep Day 4

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# Outline

- 1 Linear Algebra Problems
  - Basic problems

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# Problem 1

If the matrices

$$\begin{pmatrix} 3 & -2 & -2 \\ 1 & -1 & -1 \\ 3 & -1 & -2 \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 1 & a & 0 \\ -1 & b & 1 \\ 2 & c & -1 \end{pmatrix}$$

are inverses of each other, what is the value of  $a, b$ , and  $c$ ?

## Problem 2

In an homogeneous system of 5 linear equations with 7 unknowns, the rank of the coefficient matrix is 4. What is the maximum number of independent solution vectors?

# Problem 3

If  $A$  is a square matrix of order  $n \geq 4$ , and  $a_{ij} = i + j$  represents the entry in row  $i$  and column  $j$ , then the rank of  $A$  is always

# Problem 4

Given that  $S$  and  $T$  are subspaces of a vector space, which of the following are also subspaces?

$$S \cap T, \quad S \cup T, \quad 2S.$$

# Problem 5

If  $T$  is a linear transformation mapping vectors  $(1, 0, 0)$ ,  $(0, 1, 0)$  and  $(0, 0, 1)$  to the vectors  $(1, 2, 3)$ ,  $(2, 3, 1)$  and  $(1, 1, -2)$  respectively what vector is the image of the vector  $(3, -2, 1)$  under  $T$  ?



# Problem 6

Define linear operator  $S$  and  $T$  on the  $xy$ -plane ( $\mathbb{R}^2$ ) as follows:  $S$  rotates each vector 90 degrees counterclockwise, and  $T$  reflects each vector through the  $y$ -axis. If  $ST$  and  $TS$  denote the compositions  $S \circ T$  and  $T \circ S$ , respectively, and  $I$  is the identity map, which of the following statements is true?

$$ST = I, \quad ST = -I, \quad TS = I, \quad TS = ST, \quad TS = -ST$$

# Problem 7

Let  $T : \mathbb{R}^5 \rightarrow \mathbb{R}^3$  be a linear transformation whose kernel is a three dimensional subspace of  $\mathbb{R}^5$ . Then is  $\{T(x) : x \in \mathbb{R}^5\}$  just the origin, or a line through the origin, or a plane through the origin, all of  $\mathbb{R}^3$ , or something else? Is it possible to tell?

# Problem 8

Let  $A$ ,  $B$  and  $C$  be real  $2 \times 2$  matrices, and let  $0$  denote the  $2 \times 2$  zero matrix. Which of the following statements is/are true?

- (a)  $A^2 = 0$  implies  $A = 0$
- (b)  $AB = AC$  implies  $B = C$
- (c)  $A$  is invertible and  $A = A^{-1}$  implies  $A = I$  or  $A = -I$ ?

# Problem 9

If  $V, W$  are 2-dimensional subspaces of  $\mathbb{R}^4$ , then what are the possible dimensions of  $V \cap W$ ?

# Problem 10

Suppose that  $V$  is the vector space of real  $2 \times 3$  matrices. If  $T$  is a linear transformation from  $V$  onto  $\mathbb{R}^4$  (ie. surjective!), then what is the dimension of the null space of  $T$ ?